

Requirements Table

Telemedicine Slot Booking & Prescription Portal

Actors: Patient, Attending Physician, Payment Gateway (external system).

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Set
FR-001	Functional	The system shall enable physicians to author and digitally sign structured prescriptions that are automatically compiled into downloadable PDF records.	High	Pass: Valid cryptographic signature attached to prescription PDF. Fail: Prescription generated without doctor license validation.	Ensures prescriptions are legally valid, tamper-evident, and traceable to a licensed physician.	Set 4
FR-002	Functional	The system shall allow a patient to search and filter available physicians by medical specialty and real-time slot availability.	High	Pass: Results show only physicians with ≥ 1 open slot in the next 7 days, correctly filtered by specialty. Fail: Results include unavailable or wrong-specialty physicians.	Core discovery function patients need before booking; reduces failed booking attempts.	Set 1
FR-003	Functional	The system shall allow a patient to book a specific video consultation slot, reserving it exclusively for that patient once payment is confirmed.	High	Pass: The booked slot disappears from all other patients' availability lists within 2 seconds. Fail: The same slot can be booked by two different patients.	Prevents double-booking and guarantees the integrity of the appointment schedule.	Sets 1, 2
FR-004	Functional	The system shall automatically generate and transmit a unique, encrypted video-consultation room link to both patient and physician no later than 10 minutes before the scheduled appointment.	High	Pass: Both parties receive a working, single-use encrypted link ≥ 10 minutes before start time. Fail: Link is missing, reused across sessions, or delivered late.	Ensures secure, timely, and private access to the tele-consultation room.	Set 3
FR-005	Functional	The system shall allow a patient to download their digitally signed prescription as a PDF from their account within 24 hours of consultation completion.	Medium	Pass: Patient downloads a valid, signed PDF from the dashboard within 24 hours of consultation end. Fail: PDF is unavailable, unsigned, or late.	Gives patients timely access to prescriptions required for pharmacy fulfillment.	Set 4
NFR-001	Nonfunctional (Performance & Security)	All video-session metadata and patient clinical notes must be encrypted in transit using TLS 1.3 and at rest using AES-256.	High	Pass: Benchmarking confirms target latency and security standards under simulated peak load. Fail: Any data found unencrypted in transit or at rest.	Protects sensitive patient health data (PHI) and satisfies healthcare data-protection obligations.	All sets
NFR-002	Nonfunctional (Performance & Scalability)	The system shall support at least 500 concurrent video-consultation sessions with end-to-end video latency under 300 ms.	High	Pass: Load test at 500 concurrent sessions shows average latency < 300 ms with zero session drops. Fail: Latency exceeds 300 ms or sessions fail under peak load.	Guarantees usable, real-time video quality during peak clinic hours.	Set 3
NFR-003	Nonfunctional (Availability & Reliability)	The system shall maintain at least 99.9% uptime during published clinic hours, with no confirmed booking or signed prescription lost during a failover.	High	Pass: 30-day monitoring shows ≤ 43 min downtime and a failover drill loses zero records. Fail: Uptime below 99.9% or any record lost.	Missed consultations and lost prescriptions carry direct clinical risk.	All sets

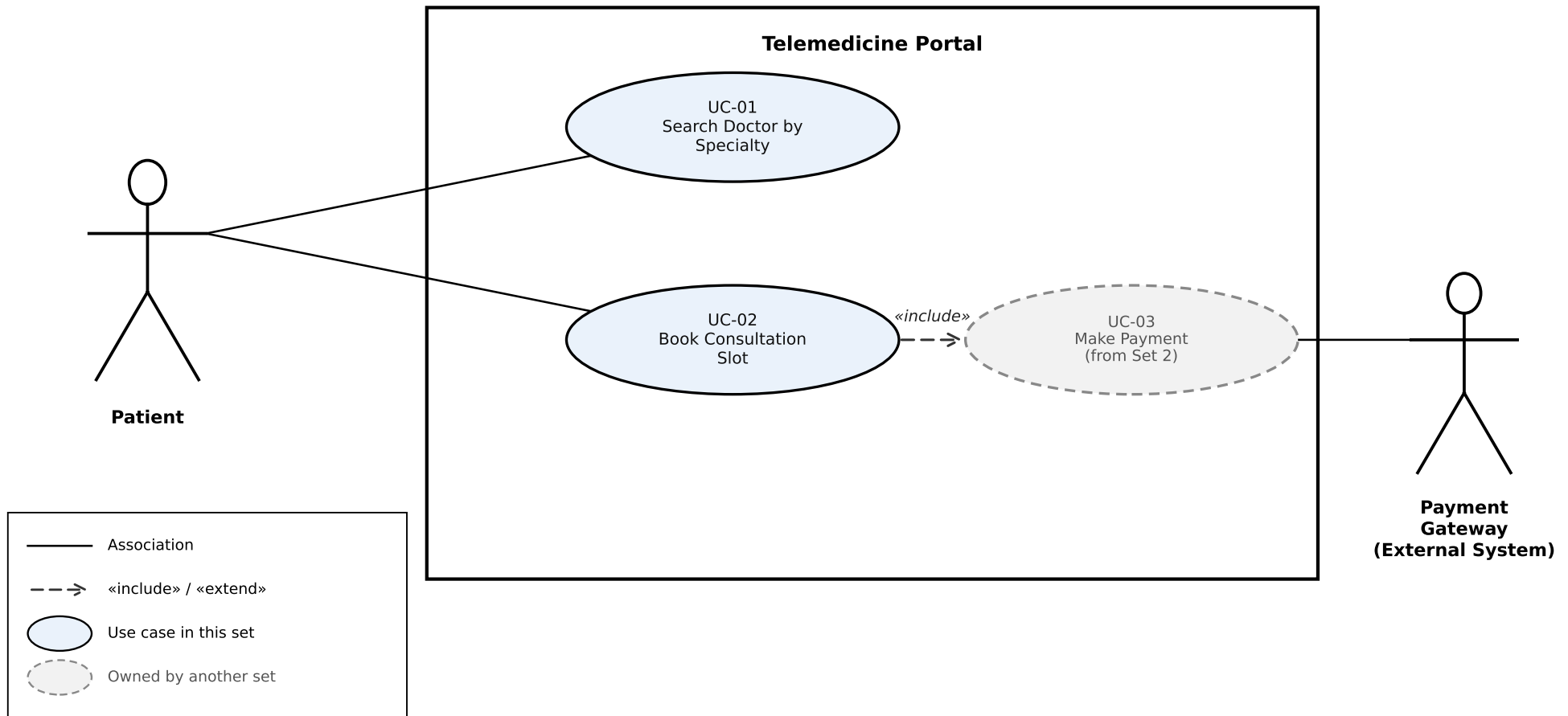
Module Split & Weightage

The eight use cases are divided into four equally weighted sets. Each set has its own use-case diagram and its own one-page use-case flow specification.

Set	Module Name	Use Cases Owned	Relationship Modelled	Weightage
Set 1	Doctor Discovery & Slot Booking	UC-01 Search Doctor by Specialty; UC-02 Book Consultation Slot	UC-02 «include» UC-03	25%
Set 2	Payment & Booking Changes	UC-03 Make Payment; UC-08 Cancel / Reschedule Slot	UC-02 «include» UC-03; UC-08 «extend» UC-02	25%
Set 3	Tele-Consultation Session	UC-04 Generate Encrypted Video Link; UC-05 Conduct Video Consultation	UC-05 «include» UC-04	25%
Set 4	Prescription Lifecycle	UC-06 Author & Digitally Sign Prescription; UC-07 Download Prescription PDF	UC-06 «extend» UC-05	25%

Set 1 — Doctor Discovery & Slot Booking

Weightage: 25% | Use cases: UC-01, UC-02



Use-Case Flow Specification

Telemedicine Slot Booking & Prescription Portal

Set 1 — Doctor Discovery & Slot Booking | Weightage: 25%

Use Case: UC-02 — Book Consultation Slot *(includes UC-03: Make Payment, owned by Set 2)*

Primary Actor

Patient

Preconditions

- Patient has a registered, authenticated account on the portal.
- Patient has searched and selected a physician by specialty (UC-01).
- At least one open consultation slot exists for the selected physician.

Postconditions

- Success: The slot is reserved exclusively for the patient, payment is captured, and a confirmation with appointment details is recorded and sent.
- Failure: No slot is reserved and no payment is captured; the patient is returned to slot selection.

Main Success Scenario

1. Patient selects an available date/time slot for the chosen physician from the booking calendar.
2. System displays a booking summary (physician, specialty, date, time, consultation fee).
3. Patient confirms the booking request.
4. System temporarily holds the slot and invokes UC-03 (Make Payment); the patient selects a payment method and submits payment.
5. System, via the Payment Gateway, authorizes and confirms the payment.
6. System permanently reserves the slot and marks it unavailable to all other patients.
7. System generates a booking confirmation with a reference ID and sends it to the patient by email/SMS.
8. System schedules encrypted video-link generation (UC-04, Set 3) for delivery before the appointment.
9. Use case ends successfully.

Alternate Flow — 5a. Payment Declined

- a. 5a1. System displays a “payment failed” message and warns that the temporary hold will expire.
- b. 5a2. Patient is prompted to retry with the same or a different payment method.
- c. 5a3. If payment fails again after 2 total attempts, the hold is released, the slot returns to the available pool, and the patient is notified that the booking was cancelled.